Articles in PresS. Am J Physiol Endocrinol Metab (May 12, 2015). doi:10.1152/ajpendo.00050.2015

Ribosome biogenesis adaptation in resistance training-induced human skeletal muscle hypertrophy.

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Running Title: Resistance training induces ribosome biogenesis.

Author Contributions: A.J.B., D.C.S., and V.C.F., provided the conception and design of the experiments. A.J.B, D.C.S., M.K.C., V.M., J.F.M. and V.C.F., were responsible for the collection, analysis and interpretation of data. V.C.F drafted the manuscript with input from all authors. V.C.F., M.K.C., J.F.M., D.C.S and A.J.B. revised and edited the manuscript. All authors approved the final version of the manuscript.

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Abstract

Resistance training (RT) promotes increases in skeletal muscle mass, with current understanding primarily implicating transient increases in the rate of muscle protein synthesis during post-exercise recovery. In contrast, the role of adaptive changes in ribosome biogenesis to support the increased muscle protein synthetic demands is not known. This study examined the effect of both a single acute bout of resistance exercise (RE) and a chronic RT program on the muscle ribosome biogenesis response. Fourteen healthy young men performed a single bout of RE, both before and after 8 weeks of chronic RT. Muscle cross-sectional area was increased by 6% ± 4.5% in response to 8 weeks RT. Acute RE-induced activation of the ERK and mTOR pathways were similar before and after RT as assessed by phosphorylation of ERK, MNK1, p70S6K and S6 ribosomal protein one hour post-exercise. Phosphorylation of TIF-IA was also similarly elevated following both RE sessions. Cyclin D1 protein levels, which appeared to be regulated at the translational rather than transcriptional level, were acutely increased after RE. UBF was the only protein found to be highly phosphorylated at rest after 8 weeks of training. Also, muscle levels of the rRNAs including the precursor 45S and the mature transcripts (28S, 18S and 5.8S) were increased in response to RT. We propose that ribosome biogenesis is an important, yet overlooked, event in resistance exercise-induced muscle hypertrophy that warrants further investigation.

Key words: ribosomal RNA, Cyclin D1, UBF, TIF-IA.

Introduction

Resistance training (RT) increases both muscle mass and strength. Increased intramuscular protein synthesis in the hours following each resistance exercise (RE) bout appears to be a major mechanism regulating the adaptive mass gain (1). This increased rate of protein synthesis results from a coordinated series of cellular events dictated acutely by an increased translational efficiency (protein synthesized per ribosome) (1, 28). The initiation of ribosomal activity, i.e., mRNA translation, is dependent on the activation of the mammalian Target of Rapamycin (mTOR) and extracellular signal-regulated kinase (ERK) pathways (8, 39). There is considerable data on the impact of exercise intensity, duration, prior training and nutritional status on these signaling pathways of translation initiation (1, 6, 28). However, a further, and less well understood, aspect of RT-induced muscle hypertrophy, is the potential for greater translational capacity (quantity of ribosome) (19). In fact, ribosomal content dictates the upper limit of protein synthesis of a cell (12), thus it may be speculated that RE not only activates translational initiation but also increases the total muscle capacity for protein synthesis through *de novo* ribosome biogenesis.

Ribosome biogenesis is a multistep process compromising ribosomal (r)RNA synthesis, processing and assembly with ribosomal proteins. The first and key limiting step is the transcription of ribosomal (r)DNA to the pre-rRNA 45S transcript by the RNA Polymerase I (Pol I) (7, 17). The 45S rRNA is subsequently cleaved, with external and internal transcribed spacers (ETS and ITS) being removed to generate three mature rRNA transcripts; 28S, 18S and 5.8S (23). These are exported to the nucleus where along with 5S rRNA (a transcript from Pol III activity) they collectively associate with a number of ribosomal proteins. The assembled mature ribosome is a complex consisting of both rRNA and ribosomal proteins distributed in two subunits: a large 60S subunit (composed of 28S,

5.8S and 5S rRNAs plus ~49 proteins) and small 40S subunit (composed of 18S rRNA plus ~33 proteins) (21, 41).

Synthesis of 45S rRNA requires the formation and activity of a complex of proteins at the rDNA promoter, including the upstream binding protein (UBF) and selectivity factor (SL1). Activation of this complex is followed by the recruitment of transcription initiation factor 1A (TIF-IA) which interacts with Pol I to drive 45S rRNA synthesis (7, 20). UBF protein content and phosphorylation state are closely correlated with Pol I recruitment and total rDNA transcription rates (31, 37). The phosphorylation of UBF at Ser388 and Ser484 is markedly heightened during cell cycle progression by the cyclins (including cyclin D1) and cyclin-dependent kinases (CDK), which are required for UBF activity (36, 37).

It has been demonstrated that ribosome biogenesis is an important component in rodent models of overload-induced muscle hypertrophy by synergistic ablation (9, 38). However, to date no study has been performed in a physiologically relevant milieu such as resistance exercise/training in human subjects. Therefore, the purpose of the present study was to examine the mechanisms of muscle ribosome biogenesis in response to a single bout of acute resistance exercise (RE) as well as chronically following a period of resistance training (RT) in healthy young adults. To achieve this we aimed to measure the signaling transduction pathways that lead to the activation of UBF and TIF-IA, the key transcriptional regulators of rRNA. Subsequently we also analyzed the impact of the resulting rDNA transcription including measurement of the precursor transcript (45S pre-rRNA), pre-rRNA sequences spanning the intermediary 5' internal/external transcribed spacer regions (ITS-28S, ETS-18S and ITS-5.8S) and the final mature rRNA transcripts (18S, 5.8S and 28S).

Materials and Methods

The study protocol was approved by the Edith Cowan University Human Ethics Committee, and was conducted in accordance with the Declaration of Helsinki. An overview of the experimental study design is shown in Figure 1.

Subjects: Fourteen males (18–36 years old) volunteered for the study, following informed written consent. For 3 weeks prior to study commencement, all participants refrained from intense lifting or strength training.

Pre-training testing: The pre-training period included two familiarization sessions: (1) an information session where proper nutrition and lifting techniques were discussed and (2) a practical familiarization session. Three days after the familiarization sessions, subjects commenced pre-training testing, which was conducted over three separate days. On day 1, total, lean and fat masses were determined using DXA scanning and thigh muscle cross-sectional area (CSA) was measured using computed tomography (CT) scanning (details below). On day 2, one-repetition maximum (1-RM) lifting performance was evaluated on the leg press, leg extension and leg flexion exercises (details below). On day 3, which was completed 4-6 days after the 1-RM testing, muscle biopsy samples were obtained from the right *vastus lateralis* both before and 1 hour after the completion of a single resistance exercise session. One hour post exercise was chosen because it is known to promote great response on both ERK and mTOR pathways, thus we hypothesize that it would also allow us to observe differences in the ribosome biogenesis markers that are dependent on these pathways.

Body composition: Total, lean and fat masses of the whole body were measured using a Hologic Discovery QDR-4500 dual energy x-ray absorptiometry (DXA) scanner (Hologic Inc., Bedford, MA, USA). Lean and fat masses (g) were calculated from total and regional analyses of the whole body scan, which provided descriptive data of the subjects and also allowed pre-biopsy and post-training session nutrition to be set (details below). CT scans

were taken of the right thigh at 50% of the distance from the lateral condyle of the femur to the greater trochanter (i.e. thigh length) by a qualified radiographer to determine quadriceps muscle CSA using Siemens SOMATOM Dual Source 64 Slice CT system (Siemens AG, Berlin, Germany). Analysis was performed using ImageJ with the mean of CSA score of two separate traces being used (33). After 8 weeks of training, CT scanning was completed 6-7 days after the last session for post-training CSA measurement. The typical error for repeated digitization of 14 scans was 1.13 cm^2 (90% CI = 0.86 – 1.68 cm^2), and the coefficient of variation was 1.2% (90% CI = 0.9 – 1.7%).

1-RM testing: Each subject's maximum lifting performance was determined on the leg press, leg extension and leg flexion (leg curl) exercises. After a full warm-up including 5 min of stationary cycling and two practice sets (6 repetitions at loads equal to ~50 and 70% of perceived maximum load), each subject performed single repetitions of an incrementally greater load until a maximum was found. Subjects typically required 4 – 6 lifts to achieve maximum. The rest interval for 1-RM attempts was 3 min. These 1-RM scores were used to monitor changes in strength performance with training and to set the training loads.

Acute resistance exercise (RE) bout and muscle biopsy procedure: Subjects reported to the lab after an overnight fast. They consumed a standardized meal (cereal plus milk) containing 2220 kJ energy, 11.6 g protein, 105 g carbohydrate and 5.6 g fat per 100 kg of lean mass. The same meal was provided for the post-training acute exercise bout and biopsy sampling session. After resting quietly for 1 h, muscle biopsy samples were taken from the right *vastus lateralis* (untrained rest biopsy) at 50% of the distance from the lateral femoral condyle to the greater trochanter (i.e. at the same thigh location as the CT scan) using the micro-biopsy technique. A 13-gauge catheter was inserted into the muscle under local anesthetic (5% lidocaine cream), then a 14-gauge triggered microbiopsy needle was inserted through the catheter. Three small (10–30 mg) samples were taken, with the catheter remaining inserted

148 for the duration. Subjects remained resting in the supine position for 30 min until warm-up
149 for the RE session commenced. The acute RE test session was the same stimulus as that used
150 throughout the RT program described in detail below. Biopsy samples were again taken 1 h
151 after the completion of the RE session (untrained post-exercise biopsy). Each sample was
152 immediately frozen in liquid nitrogen and stored at -80°C until further analysis. After 8
153 weeks of training, the muscle biopsy sampling procedure was repeated 3-5 days after the last
154 training session (trained rest biopsy, and 1 h after RE, trained post-exercise biopsy). Subjects
155 consumed a whey protein-laden drink (RedBak® Whey Protein 50/50 isolate/concentrate,
156 Probiotec Limited, VIC, Australia) immediately after every RE session throughout the
157 training program and both RE testing sessions. The drink contained 38.4 g protein, 3.8 g
158 carbohydrate and 2.1 g fat per 100 kg lean mass; thus, post-exercise nutrition was identical in
159 training and testing sessions.

160 *Resistance training (RT) program:* Subjects were paired for 1-RM knee extension strength
161 and then randomly allocated to one of two training groups: heavy-load group training at 90%
162 of 1-RM (90% 1-RM group) or a moderate-load group training at 70% of 1-RM (70% 1-RM
163 group) for leg press, knee extension and knee flexion exercises. Each subject was supervised
164 at each session by an experienced strength trainer who was able to motivate the subjects and
165 ensure that they could safely complete each set to absolute failure. The subjects performed as
166 many repetitions as possible in the first set of the first session of testing. The number of
167 repetitions was recorded and then loads were adjusted set-by-set during training to maintain
168 the number of repetitions; thus loads were continually increased as the subjects increased
169 their strength. Each subject's 1-RM load was determined before and after 4 weeks of training
170 to allow for adjustment of the repetition range if the load-repetition relationship changed.
171 Training was performed twice a week for 8 weeks (3 exercises, four sets each until failure),
172 with 2 min rest between sets and 3 min rest between exercises.

173 *Analysis of muscle tissue*: *Total RNA extraction and cDNA synthesis*: Total RNA was
174 extracted using the ToTALLY RNA Kit (Ambion Inc., Austin, TX) following manufacturer's
175 instructions. RNA integrity was determined using the Agilent 2100 BioAnalyzer (Agilent
176 Technologies, Mulgrave, VIC, Australia). RNA samples were then diluted to an appropriate
177 concentration with nuclease-free water (NFW) and cDNA synthesis performed using the
178 High Capacity RNA-to-cDNA kit (Applied Biosystems, Foster City, CA, US). cDNA was
179 stored at -20°C for subsequent analysis.

180 *Real-Time PCR:* RT-PCR was performed on a LightCycler 480 II using SYBR Green I
181 Master Mix (Roche Applied Science, Penzberg, Germany). Commonly used reference genes
182 hypoxanthine-guanine phosphoribosyltransferase (HPRT), TATA-binding protein (TBP),
183 heat shock protein 90 (HSP90), peptidyl-prolylcis-trans isomerase (PPIA also known as
184 cyclophilin A), beta-2 microglobulin (B2M) and glyceraldehyde 3-phosphate dehydrogenase
185 (GAPDH) mRNA levels were measured (Table 1). Geometrical means of the three most
186 stable and with lower variance (based on the technique described by Mane *et al.* (18))
187 mRNAs (HPRT, TBP and PPIA) were calculated and used for normalisation of the target
188 RNAs, by the 2(-ΔΔCT) method. Target mRNAs were: UBF, polymerase RNA I polypeptide
189 B (POLR1B), cyclin D1, TIF-IA (also known as RRN3). Target rRNAs were the mature
190 ribosome species 28S, 18S, 5.8S. Since these primers should amplify both mature rRNAs but
191 also pre-rRNAs, primers were designed specifically for pre-rRNAs sequences spanning the 5'
192 end internal/external transcribed spacer region (ETS and ITS respectively) and a specific
193 primer for the initial region of 5' end of 45S rRNA, namely 45S, ETS-18S, ITS-5.8S and
194 ITS-28S (Fig 2). Primers for rRNAs (Table 1) were designed by QIAGEN specific for this
195 study, using RT^2 Profiler PCR Arrays (QIAGEN, Venlo, Limburg, Netherlands). Standard
196 and melting curves were performed for every target to confirm primer efficiency and single
197 product amplification.

198 *Western blot*: Skeletal muscle tissue was homogenized using an OMNI Bead Ruptor
199 Homogenizer (Omni International, Kennesaw, GA, US) using RIPA lysis buffer (Tris 50 mM
200 Tris-HCl, pH 7.4, 150 mM NaCl, 1% deoxycholic acid, 1% NP-40, 1mM EDTA)
201 supplemented with Halt™ Protease and Phosphatase Inhibitor Cocktail (Thermo Scientific,
202 Waltham, MA, USA). After a centrifugation step, the supernatant had the protein
203 concentration determined with the BCA protein assay kit (Pierce, Rockford, IL, USA).
204 Protein homogenates were mixed with 4× Laemmli's buffer and boiled. Denatured proteins
205 were separated by SDS-PAGE using 8 to 18% gel depending on the protein target. Proteins
206 were electrotransferred to PVDF membranes (BioRad, Hercules, CA, US) using TransBlot
207 semi-dry transfer apparatus (BioRad). Gels were cut and transferred together, as described
208 elsewhere (15). This procedure allows different proteins to be quantified in the same run
209 avoiding multiple stripping and reprobing cycles plus saving sample material. Membranes
210 were blocked for 1 h in 5% BSA in TBST solution at room temperature and probed using
211 specific antibodies for pan cyclin D1, pan UBF, p-UBF Ser388, p-UBF Ser484 (Santa Cruz
212 Biotechnology, Dallas, TX, US), p-ribosomal protein S6 Kinase, 70kDa, polypeptide 1
213 (p70S6k) Thr421/Ser424, p-p70S6k Thr389, p-ERK 1/2 Thr202/Tyr204 and Thr185/Tyr187,
214 p-mitogen-activated protein kinase interacting kinase 1 (MNK1) Thr197/202, p-S6 ribosomal
215 protein Ser240/244, p-S6 ribosomal protein Ser235/236, p-eIF4E Ser209, pan eIF4E-binding
216 protein 1 (4E-BP1) (Cell Signaling Technologies, Danvers, MA, USA), pan TIF-IA, p-TIF-
217 IA Ser649 and pan GAPDH (Abcam, Cambridge, MA, US). Primary antibodies were
218 incubated overnight in blocking buffer (1:1,000 dilution, with exception of GAPDH,
219 1:10,000) followed by the appropriate anti-rabbit or anti-mouse linked to horseradish
220 peroxidase secondary antibody (1:5,000) for 1 h at room temperature at the subsequent day.
221 Membranes were exposed using enhanced chemiluminescence reagent (ECL Select kit, GE

HealthCare) and bands were detected using ChemiDoc (BioRad). Bands were quantified using ImageJ software (NIH, Bethesda, US). Western blot data were normalized by GAPDH.

Statistics: Data are expressed as means ± standard error of mean (SEM), unless otherwise noted. Quadriceps CSA and percentage changes in quadriceps CSA (% ΔCSA) from pre-training were analyzed by two-way ANOVA with repeated measures (group [intensity] × time). Baseline anthropometric characteristics were analyzed by student's t-tests. Western blot data are presented as fold change against the pre-training rest biopsy sample. The effects of acute resistance exercise (RE) and chronic resistance training (RT) were assessed using a two-way ANOVA with repeated measures (SigmaPlotv12.0). Following statistically significant main effects, Student-Newman-Keuls post-hoc tests were used to determine the significance of pair-wise comparisons between individual acute resistance exercise sessions and training. For the scope of the current study, the effects of 'resistance exercise' is defined as the acute response of a single session of exercise (i.e. rest versus post-RE), and the effects of 'resistance training' refers to the chronic response after 8 weeks of training (i.e. untrained versus trained) both at rest (untrained rest versus trained rest) and the response induced by exercise over time (untrained post-RE versus trained post-RE). When appropriate, correlations between variables were assessed using Pearson's product moment correlations. Statistical significance was accepted at $p \leq 0.05$.

Results

Subject characteristics

The subject's baseline anthropometric characteristics are shown in Table 2.

Muscle cross-sectional area (% ΔCSA)

245 Mid-thigh muscle CSA increased significantly following the 8-week RT program (6.4±2.0
246 and 7.6±1.0%, 70 and 90% 1-RM groups respectively, $P<0.05$ and $P<0.01$), with no
247 difference between groups ($P=0.56$) (Fig 3). Thus, subsequent molecular analysis either
248 examined the pooled group means or each subject's individual data in cases of correlations
249 between changes in CSA and measures of ribosome biogenesis.

250 ***Effect of resistance exercise and training on translation initiation pathways (ERK and***
251 ***mTOR)***

252 *p70S6K*: Phosphorylation of p70S6K at Thr389 (Fig 4A) increased post-RE (main effect,
253 $P<0.001$) by 6-fold in untrained ($P<0.05$) and 6.5-fold in trained conditions ($P<0.05$). There
254 was also a main effect of RE for p70S6K phosphorylation at Thr421/Ser424 ($P<0.001$) (Fig
255 4B), with increases in response to exercise of 2.3-fold in untrained (trend, $P=0.052$) and 2.9-
256 fold in trained conditions ($P<0.05$). There was no effect of training for either p70S6K
257 phosphorylation site.

258 *ERK 1/2:* Phosphorylation of ERK1 (Thr202/Tyr204) (Fig 4C) and ERK2 (Thr185/Tyr187)
259 (Fig 4D) showed a main effect of exercise (both $P<0.001$) but no effect of training. ERK1
260 phosphorylation was increased 3.2-fold post-RE in the untrained condition ($P<0.05$), and 2.9-
261 fold in the trained condition (trend, $P=0.064$). RE-induced ERK2 phosphorylation was
262 increased 2.2-fold before training ($P<0.01$), and 2.1-fold after training ($P<0.05$).

263 *S6 ribosomal protein*: Phosphorylation of S6rp at Ser235/236 (Fig 4E) was increased after
264 exercise (main effect, $P<0.01$). Before training, RE resulted in a 4.5-fold change ($P<0.05$),
265 whereas after training, a 4.2-fold change was observed ($P<0.05$). There was also a main
266 effect for exercise ($P<0.001$) for phosphorylation of Ser240/244 (Fig 4F) with an 8.2-fold
267 elevation pre-training ($P<0.05$) and 6-fold elevation post-training ($P<0.05$). There was no
268 significant effect of training.

MNK1: Phosphorylation at Thr197/202 (Fig 4G) was increased markedly in response to RE (main effect, $P<0.001$, no effect of training); which was 7-fold both in untrained ($P<0.01$) and trained conditions ($P<0.01$).

eIF4E and 4E-BP1: There was no statistical difference acutely nor chronically for eIF4E phosphorylation at Ser209 or hyperphosphorylation of 4E-BP1 assessed by mobility shift (data not shown).

Effect of resistance exercise and training on cell cycle regulators

Cyclins: Cyclin D1 protein (Fig 5A) increased 1.9-fold after both bouts of RE (main effect $P<0.01$, before and after training, both $P<0.01$), and there was no training effect. There was no difference in cyclin D3 protein (Fig 5B) at any time point. Cyclin D1 mRNA (Fig 5D) showed a training × exercise interaction ($P<0.001$) and a main effect for exercise ($P<0.05$). This was due to a small, but significant, decrease in resting cyclin D1 mRNA following training ($P<0.05$) together with an acute increase in mRNA post-exercise in the trained condition only ($P<0.05$).

Effect of resistance exercise and training on UBF and TIF-IA

Total and phosphorylated UBF: UBF total protein was markedly increased after exercise (main effect, $P<0.01$, although as discussed later, we believe this result is likely be an artefact from protein extraction), but was not influenced by resistance training (Fig 6A). Substantial variation in subject responsiveness to acute exercise was noted for total UBF. Total UBF protein levels were negatively correlated with UBF phosphorylation at Ser484 ($r=-0.54$, $P<0.001$) and Ser388 ($r=-0.32$, $P<0.05$, data not shown).

There was no effect of exercise or training on phosphorylation of UBF at Ser484 (Fig 6C). Additionally, there was no effect of acute exercise on the phosphorylation of UBF at Ser388

(Fig 6B). However, UBF phosphorylation at Ser388 was chronically increased by 5 fold in resting muscle following RT (P<0.05) (Fig 6B).

TIF-IA: Phosphorylation at Ser649 of TIF-IA (Fig 6D) was markedly increased after RE in both conditions (main effect, P<0.01, 5.6-fold change before, P<0.01; and 7.6-fold after RT, P<0.01). There was a trend for a greater effect of RE after the training period (P=0.06). There was no effect of RE or RT on total TIF-IA (Fig 6E).

Effect of resistance exercise and training on rRNA and total RNA

Total muscle RNA concentration (μg of RNA per mg of tissue) tended to be increased post-RT (1.3-fold), although there was considerable subject-to-subject variation and this data did not reach statistical difference (data not shown). Nevertheless, the extent of each subject's fold change in muscle total RNA from pre- to post-training showed a significant positive correlation with muscle % ΔCSA (r=0.72, P=0.01) (Fig 7A).

45S pre-rRNA (primer internal to the 5'ETS region) expression showed an interaction between resistance exercise and training (P<0.05). Post-hoc analysis revealed an increase in 45S expression in resting muscle following RT (>2 fold increase, P<0.01, Fig 7B). After RT, but not before RT, there was an acute RE-induced decrease in 45S rRNA expression (P<0.01) (Fig 7B). Levels of 45S pre-rRNA following exercise were strongly correlated with the extent of TIF-IA phosphorylation at Ser649 (r=0.66, P<0.01) but not UBF phosphorylation at Ser388 and Ser484 (r=-0.36, P =0.14; and r=0.23 and P=0.4, respectively).

There were no significant changes after RE or RT for ETS-18S ITS-5.8S or ITS-28S pre-rRNAs (Fig 7D, F and H), although, trends were observed for main effects of exercise for ETS-18S (P=0.08) and ITS-28S (P=0.055).

There were training effects for all mature rRNAs: 18S (P<0.01), 5.8S (P<0.01) and 28S (P<0.05) (Fig 7C, E and G). Post-hoc analysis showed differences between untrained and

316 trained conditions at rest for 18S (4 fold increase, $P<0.01$) and 5.8S (~4 fold increase,
317 $P<0.01$) and a similar trend for 28S (2-fold increase, $P=0.079$). Additionally, there were main
318 effects of exercise to acutely reduce levels for 18S ($P<0.05$) and 5.8S ($P<0.05$), but not for
319 28S.

320 ***Effect of resistance exercise and training on mRNA levels of pol I-associated proteins:***

321 Gene expressions of POLR1B and TIF-IA (Fig 8) were not significantly different at any time
322 point. UBF mRNA showed a significant 73% increase with training (main effect of training,
323 $P<0.01$) at rest ($P<0.01$). Acutely, there was a small decrease after RE following RT
324 ($P<0.05$), but not before RT.

325

326 **Discussion**

327 The novel finding of present study was that significant muscle hypertrophy induced
328 by 8 weeks of resistance training (RT) was associated with increased muscle expression of
329 ribosomal RNA transcripts accompanied by a robust activation of key ribosome biogenesis
330 regulatory factors. Specifically, it was demonstrated that 18S, 5.8S, 28S rRNAs and the pre-
331 rRNA 45S were increased by 2 to 4 fold in the trained muscle. Acute RE promoted
332 ERK/mTOR pathway activation, increased levels of cyclin D1 protein and elevated
333 phosphorylation of TIF-IA. Whilst UBF phosphorylation was not increased acutely post-RE,
334 it was markedly elevated on Ser388 in resting muscle post-training. These data show that key
335 indicators of ribosome biogenesis are responsive to a single bout of RE, whilst repeated
336 training appears to be necessary for the sustained activation of UBF and accretion of muscle
337 rRNA.

338 Signaling transduction via the ERK and mTOR pathways was greatly upregulated 1 h
339 following an acute bout of RE, both in the untrained (i.e., pre-RT) and trained (post-RT)

conditions. Phosphorylation of p70S6K at Thr389 (mTOR dependent) and Thr421/Ser424 (predominantly ERK pathway dependent) (13), as well as S6rp at Ser240/244 (p70S6K dependent) and Ser235/236 (both p70S6k and p90 ribosomal S6 kinase, RSK, dependent) (30) were greatly increased following RE. Phosphorylation of ERK 1/2 at Thr202/Tyr204 and Thr185/Tyr187, respectively, and its downstream target MNK1 at Thr197/202 were also greatly increased. However, none of these kinase responses were altered after 8 weeks of training. Our results do not corroborate the view that chronic training causes an attenuation of acute exercise-induced signal transduction pathways related to muscle hypertrophy as previously hypothesized (5, 25, 40). At least within the time frame of 8 weeks training, both acute exercise bouts resulted in a similar activation of these kinases, despite significant training induced gains in muscle mass. It remains to be determined if the activation of these kinases is diminished with ongoing training or when a plateau in training intensity and exercise adaptation occurs.

Signal transduction via the mTOR and ERK pathways enhances translational initiation and elongation of a series of specific mRNAs. Whilst it is known that ERK and mTOR are activated in the early hours of post-exercise recovery, less is known of the specific proteins that are translated due to this effect. In the present study, an increase in muscle levels of cyclin D1 protein after both acute RE bouts were observed, without concurrent increases in cyclin D1 mRNA or cyclin D3 protein. Consistently, cyclin D1 protein expression has been shown to be primarily regulated at the translational rather than transcriptional level (11, 22, 29). Although it has been hypothesized that the downstream mTOR effector 4E-BP1 may be involved in the translation of cyclin D1, it was recently shown that mTOR does not specifically regulate its expression (34). Alternatively, eIF4E was shown to be responsible for the specific translation of cyclin D1 mRNA (29), and the phosphorylation of eIF4E at Ser209 by MNK1 appears to be required for this effect (2, 35). We were not able to detect an

365 increase in eIF4E phosphorylation at Ser209 after RE, but heightened phosphorylation of the
366 two upstream kinases, MNK1 and ERK, were observed. It is possible that our single 1 h post-
367 exercise biopsy time-point may have missed an increase in eIF4E phosphorylation, since
368 cyclin D1 protein was already increased by this time. We suggest that cyclin D1 is a novel
369 molecular target which is specifically upregulated in human muscle after RE, possibly driven
370 by the ERK/MNK pathway.

371 The role of cyclin D1 in the regulation of the cell cycle/proliferation is known;
372 however, cyclin D1 also plays a key role in mature cells by promoting ribosome biogenesis.
373 Cyclin D1 forms a complex with CDK4 and promotes its catalytic activity (16), which
374 promotes subsequent phosphorylation and activation of UBF (37). Active UBF binds to the
375 rDNA promoter leading to increased rDNA transcription. In the current study,
376 phosphorylation of UBF appeared to be reduced or almost undetectable at 1 h post-RE in
377 some subjects. UBF has several HMG box domains that confers UBF a strong binding
378 affinity for rDNA (27, 42). However, once UBF is dephosphorylated, it will release rDNA
379 and potentially redistribute from the nucleolus to the nucleus or even cytoplasm (26).
380 Interestingly, those subjects who had a blunted phosphorylation of UBF after exercise also
381 displayed a robust increase in total UBF levels. We hypothesize that this effect may be due to
382 the protein extraction protocol employed in the current study. Detachment of
383 dephosphorylated UBF protein from rDNA would enrich the protein soluble fraction during
384 tissue extraction. It has been reported that even nuclear specific protein extraction protocols
385 have been found to recover only 50% of nucleolar proteins (4). It is likely that the protein
386 extraction protocol employed in this study failed to recover the entire nucleolar fraction and
387 that this accounted for the increased UBF levels after RE. Convincingly, at the same time
388 UBF was increased, TIF-IA protein levels did not show the same response post exercise (Fig.
389 6E). Although it can also be found in the nucleolar region and has been recently shown to be

able to bind to DNA in vitro (32), TIF-IA possess several HEAT domains, known to be involved in protein-protein interactions (3). Its main role on ribosome biogenesis is through these protein-protein interactions, serving as a bridge between Pol I and SL1 (10). Since protein-protein interactions are easily disrupted by ionic/denaturing lysis buffer, total levels of TIF-IA was not affected by the protein extraction protocol. Thus we suggest that the UBF increase observed after exercise is an indirect effect due to its detachment from rDNA.

Initially we hypothesized that total UBF protein would be increased chronically in hypertrophied muscle following chronic resistance training, since it has been shown to be upregulated during synergistic ablation in a mouse model of muscle hypertrophy (9, 38). We found, however, no evidence that muscle hypertrophy following 8 weeks of resistance training in humans was associated with increases in muscle UBF protein levels or Pol1 factors PolR1B and TIF-IA mRNA (despite increased UBF mRNA levels). In line with this, Zhang et al. (44) showed an increase in ribosomal RNA without concurrent increases in total UBF protein expression in cardiomyocytes. They speculated that the ribosomal RNA accretion may involve activation of the pre-existing components of Pol I and III machinery, which may also be the case here. We did find a robust increase in UBF phosphorylation at rest in trained muscle, which might help to explain the chronic increase in rRNAs. Contrary to murine models of hypertrophy (mechanical overload or surgical ablation) where there is a sharp increase in hypertrophy (40 to 70% of muscle mass increment in 1-3 weeks), resistance training produces a much slower rate of hypertrophy (~7% over 8 weeks in the current study). It is plausible that human muscle ribosome accretion with resistance training may be more dependent on the activation of pre-existing components of ribosome biogenesis pathways, including UBF phosphorylation, rather than increments in total UBF protein. Notably, UBF was the only protein measured in the current study that demonstrated an increase in phosphorylation state in trained muscle at rest. However, a few other possible explanations

exist that may explain the observed absence of chronic elevation in UBF protein in human muscle tissue. First, as mentioned above, the extraction protocol may not have been optimal for nucleolar proteins, and this may have underestimated any potential real increase in total UBF. Second, RT resulted in a much smaller increase in muscle mass compared to animal models of synergistic ablation, which may still cause a proportional increase in UBF level, but these may be below the limit of detection of the western blot technique employed. Third, one cannot rule out the possibility that the synergistic ablation model simply increases muscle mass by different mechanisms compared to the more physiological stimulus of human RT.

Unlike the chronic effect on UBF phosphorylation, TIF-IA was greatly phosphorylated at the ERK-RSK dependent site (Ser649) after acute RE, but not chronically. Although rRNA synthesis relies greatly upon UBF activation, it may, in some specific cases, be increased by TIF-IA activation by the ERK pathway, despite a lack of a change in UBF phosphorylation status (45). Along with cyclin D1 protein, TIF-IA may be an earlier activator of ribosome biogenesis after RE that may promote rRNA synthesis at a later time point.

Synthesis of rRNA has been found to be an early event in rodent models of skeletal muscle hypertrophy (9, 38). However, whether rRNA synthesis plays a similar role in resistance training-induced muscle hypertrophy in humans remains unclear. In the current study, relative levels of mature rRNAs and the pre-rRNA 45S (primer designed against the 5'ETS region not spanning any mature rRNA) were markedly increased in resting muscle following 8 weeks of resistance training. Since the pre-rRNA 45S was similarly increased, we asked whether the mature rRNA data showed higher levels due to the pre-rRNAs, since primers designed for mature rRNAs will also amplify the pre-rRNAs version. In order to differentiate pre-rRNA from mature rRNA we also analyzed the 28S, 18S and 5.8S segments which were still bound to the internal (ITS-28S and ITS-5.8S) and external transcribed spacer (ETS-18S) within the pre-rRNA 45S strand. Unlike the primers designed for the internal

region of the mature rRNA alone, none of the pre-rRNA fragments (ITS-28S, ETS-ITS and ITS-5.8S) were increased after training. The training-induced increase in pre-rRNA 45S, but not ETS-18S, ITS-5.8S or ITS-28S, may suggest that pre-rRNA is rapidly processed. Overall, our data show that resistance training induces long term increases in muscle ribosome machinery. A major implication of this finding is that rRNAs, which are commonly used as reference genes in the literature, clearly cannot be used as reference genes in longitudinal studies, especially when there are gains and/or losses of muscle mass.

Despite there being chronic increases in mature rRNA abundances post-training in the current study, we did not find an acute increase in 45S pre-rRNA levels in response to RE. For long term increases in rRNA, rDNA transcription should be initiated at some point following acute RE. We hypothesize that rDNA transcription may be upregulated at a later time point during post-exercise recovery. In line with this hypothesis, Nader et al. (24) recently reported an increase in 45S rRNA 4 h post-RE, three hours later than our biopsy sampling time-point. Convincingly, even though the levels of 45S pre-rRNA after exercise were decreased in the present study, they were positively correlated with the levels of TIF-IA phosphorylation (but not UBF phosphorylation). This may indicate that the transcription of rDNA induced by resistance exercise is mediated by ERK-RSK-TIF-IA pathway, independently of UBF.

Total RNA per mg of muscle tissue appeared to be slightly increased following chronic resistance training but this change showed a high inter-individual variability and the overall change did not reach statistical significance. However, perhaps more importantly, the extent of change in a subject's total muscle RNA concentration was found to be positively correlated with the extent of changes in muscle CSA (Fig. 7A). Total RNA is known to be a proxy measurement of ribosomal RNA content, since >80% of total RNA is ribosomal RNA

(43). Therefore, the degree to which subjects were capable of muscle mass gains appears related to their extent of increase in muscle ribosome density.

In conclusion, the current data indicate that ribosome biogenesis is a relevant event potentially driving muscle adaptation to resistance training. We propose a testable model (Fig 9) where dephosphorylated UBF is rapidly removed from rDNA promoter in the nucleolus during or shortly after contraction. The muscle cell requires energy during contraction, which will shut down high energy consumption activities such as protein synthesis and ribosome biogenesis. However, at the same time, RE sends antagonistic cues for latter adaptation, where cyclin D1 protein expression may be its earliest triggering regulator, possible by the ERK-MNK1-eIF4e pathway, and TIF-IA is activated independently of UBF. Moreover, the increase in ribosome density may explain the degree of muscle hypertrophy, i.e., reduced accretion of rRNA may limit muscle growth. Some studies have found that RT increases basal muscle protein synthesis (14, 40), which appears not to be mediated by the signaling pathways that increase protein translation efficiency. We show evidence that the increase in protein synthesis at rest might be mediated by increased ribosome biogenesis in response to resistance training.

Acknowledgements

The authors would like to thank Dr. Matthew Barnett for the critical analysis of the paper.

Grants

This work received funding from Edith Cowan University research center funding scheme (to the Centre for Exercise and Sports Science Research). V.C.F is supported by a University of Auckland Doctoral Scholarship and by AgResearch Core funding.

Conflict of Interest

The authors declare no conflict of interest.

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Figure Legends

Figure 1: Experimental study design. *A*, timeline of measurements and interventions; *B*, timeline of the acute clinical trial design used in both occasions (i.e., untrained and trained state).

Figure 2: Real Time PCR primers for rRNAs. Pre-rRNA 45S representation, black filled rectangles represent regions that will generate mature 18S, 5.8S and 28S, with ETS (external transcribed spacer) and ITS (internal transcribed spacer) in between. Primers used for accessing pre-rRNAs and mature rRNAs were designed against the regions depicted and shown as lowercase letters: a, internal to ETS, specific for 45S; b, ETS-18S; c, internal to 18S; d, ITS-5.8S; e, internal to 5.8S; f, ITS-28S; g, internal to 28S.

Figure 3: Resistance training increased quadriceps muscle cross-sectional area similarly in both 70% and 90% 1-RM intensity groups. Percentage increase in cross-sectional area per group; data are presented as mean values with maximum and minimum variation**.** Both intensities of training promoted statistically similar gains.

Figure 4: Effect of resistance exercise and training on a panel of kinases from ERK and mTOR pathway. *A*, phosphorylation of p70S6K at Thr389; *B*, at Thr421/Ser424; *C*, phosphorylation of ERK1 at Thr202/Tyr204; *D*, ERK2 at Thr185/Tyr187; *E*, phosphorylation of S6rp at Ser235/236; *F,* at Ser240/244; *G*, Phosphorylation of MNK1 at Thr197/202; *H,* representative western blots. All western blot data were normalized to GAPDH. Values are mean ± SEM. * $p \leq 0.05$, ** $p \leq 0.01$, versus respective baseline; # $p \leq 0.05$, versus the same

exercise state (rest or post-RE [post]) of the opposite training condition (untrained or trained). Main effects and interactions are presented in the text.

Figure 5: Effect of resistance exercise and training on cell cycle regulators. *A*, levels of cyclin D1 protein; *B*, cyclin D3 protein; *C*, representative western blots. Western blot data were normalized to GAPDH; *D*, cyclin D1 mRNA expression normalized by geometrical mean of three reference genes. Values are mean ± SEM. * $p \leq 0.05$, ** $p \leq 0.01$ versus respective baseline; # $p \leq 0.05$, versus the same exercise state (rest or post-RE [post]) of the opposite training condition (untrained versus trained). Main effects and interaction are presented in the text.

Figure 6: Effect of resistance exercise and training on total and phosphorylation of UBF and TIF-IA. *A*, UBF content before and after training and exercise; *B*, phosphorylation of UBF at Ser388; *C*, at Ser484; *D*, phosphorylation of TIF-IA at Ser649*; E*, TIF-IA content before and efter training and exercise; *F*, Representative western blot figures for UBF and TIF-IA phosphorylation residues and total protein leves, and GAPDH are shown. Western blot data were normalized to GAPDH. Values are mean ± SEM. * $p \leq 0.05$, ** $p \leq 0.01$, versus respective baseline; # $p \leq 0.05$, versus the same exercise state (rest or post-RE [post]) of the opposite training condition (untrained or trained). Main effects and interactions are presented in the text.

Figure 7. ***Effect of resistance exercise and training on total RNA, mature and pre-rRNA levels.*** *A*, correlation between total RNA per mg of muscle tissue and percentage increase in muscle cross-sectional area; the calculation used for total RNA measurement is described in Methods section. Due to technical problems, it was not possible to back calculate total RNA

665 for three samples; *B,* relative levels of 45S rRNA*; C,* 18S rRNA; *D*, ETS-18S pre-rRNA; *E*,
666 5.8S rRNA; *F*, ITS-5.8S pre-rRNA; *G,* 28S rRNA; *H*, ITS-28S rRNA. Data were normalized
667 to the geometric mean of three housekeeping genes (3HK). Values are mean ± SEM. *
668 p≤0.05, **p≤0.01, versus respective baseline; # p≤0.05, versus the same exercise state (rest
669 or post-RE [post]) of the opposite training condition (untrained or trained).
670
671 **Figure 8. Effect of resistance exercise and training on mRNA levels of Pol I associated**
672 **proteins.** Pol1 factors PolR1B and TIF-IA along with the associated factor UBF are shown.
673 Data were normalized to the geometric mean of three housekeeping genes (3HK). Values are
674 mean ± SEM. * p≤0.05, **p≤0.01, versus respective baseline; # p≤0.05, versus the same
675 exercise state (rest or post-RE [post]) of the opposite training condition (untrained or trained).
676
677 **Figure 9. Theoretical model for the early and delayed effects of resistance exercise in**
678 **human skeletal muscle.** Resistance exercise promotes TIF-IA phosphorylation, mTOR and
679 ERK signaling that specifically increases translation of cyclin D1. Increased cyclin D1 may
680 promote activation of UBF protein, increasing rDNA transcription and sustaining muscle
681 growth chronically. The figure depicts the nucleus and nucleolus (darker region) in the
682 center, where rDNA is concentrated.

Table 1. RT-PCR primer details.

Target	Catalog number
Span 5'-end 5.8S rRNA	PPH82111A
Internal to 5.8S rRNA	PPH82091A
Span 5'-end 18S rRNA	PPH82110A
Internal to 18S rRNA	PPH71602A
Span 5'-end 28S rRNA	PPH82112A
Internal to 28S rRNA	PPH82090A
45S rRNA (5'ETS)	PPH82089A
TATA box binding protein (TBP), Forward	TGTGCTCACCCACCAACAAT
TATA box binding protein (TBP), Reverse	TCTGCTCTGACTTTAGCACCTG
Cyclophilin A (PPIA), Forward	CCAGGCTCGTGCCGTTTTG
Cyclophilin A (PPIA), Reverse	GTCTGCAAACAGCTCAAAGGAG
Hypoxanthine phosphoribosyltransferase 1 (HPRT), Forward	CCTGGCGTCGTGATTAGTGAT
Hypoxanthine phosphoribosyltransferase 1 (HPRT), Reverse	TCGAGCAAGACGTTCAGTCC
Cyclin D1, Forward	GCTGCGAAGTGGAAACCATC
Cyclin D1, Reverse	CCTCCTTCTGCACACATTTGAA
RRN3 RNA polymerase I transcription factor (TIF-IA), Forward	GTTCGGTTTGGTGGAACTGTG
RRN3 RNA polymerase I transcription factor (TIF-IA), Reverse	TCTGGTCATCCTTTATGTCTGG
Upstream binding transcription factor (UBF), Forward	CCTGGGGAAGCAGTGGTCTC
Upstream binding transcription factor (UBF), Reverse	CCCTCCTCACTGATGTTCAGC
Polymerase (RNA) I polypeptide B (POLR1B), Forward	GCTACTGGGAATCTGCGTTCT
Polymerase (RNA) I polypeptide B (POLR1B), Reverse	CAGCGGAAATGGGAGAGGTA

Table 2. Subject baseline anthropometric characteristics.

Group	Age (years)	Height (cm)	Mass (kg)	Lean Mass (kg)
70% 1-RM	20.5 ± 1.9	181.4 ± 10.3	81.1 ± 22.1	64.6 ± 12.9
90% 1-RM	22.4 ± 5.2	177.5 ± 9.4	78.8 ±8.6	61.8 ± 5.0
All	21.6 ±4.1	179.2 ± 9.6	79.8 ± 15.1	63.0 ± 8.9
***P* value**	*0.42*	*0.48*	*0.78*	*0.56*

Data are Mean ± SD, 70% 1-RM group, n=6; 90% 1-RM, n=8.

Figure 1.

A

Weeks 1 and 2	***Week 3***	***Weeks 4-12***	***Week 13***
CT scan *DEXA scan* *1-RM test*	*Untrained* *RE session*	*8 weeks of RE* *Training* *1-RM retest* *Trained RE session*	*CT scan*

B

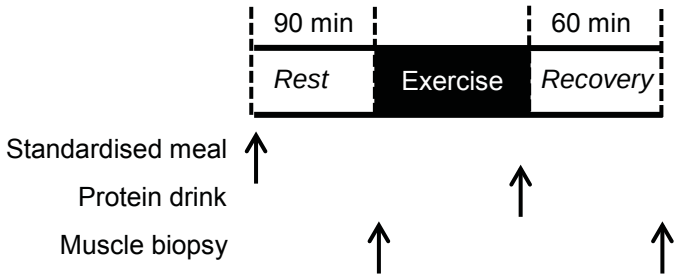


Figure 2.

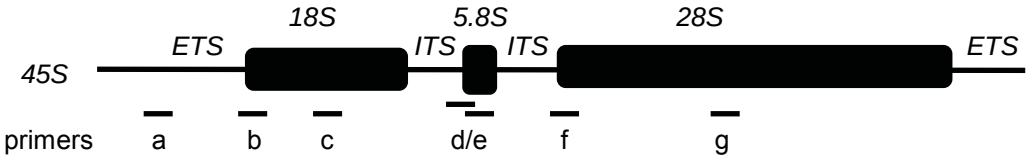


Figure 3.

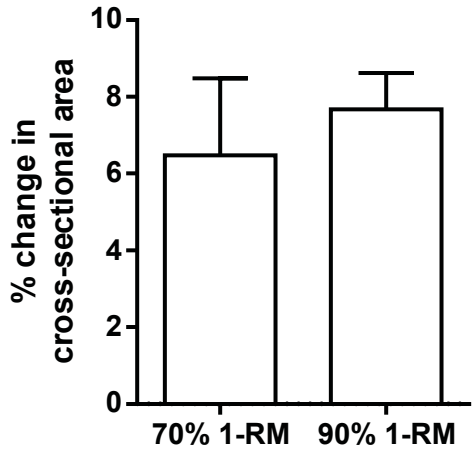


Figure 4.

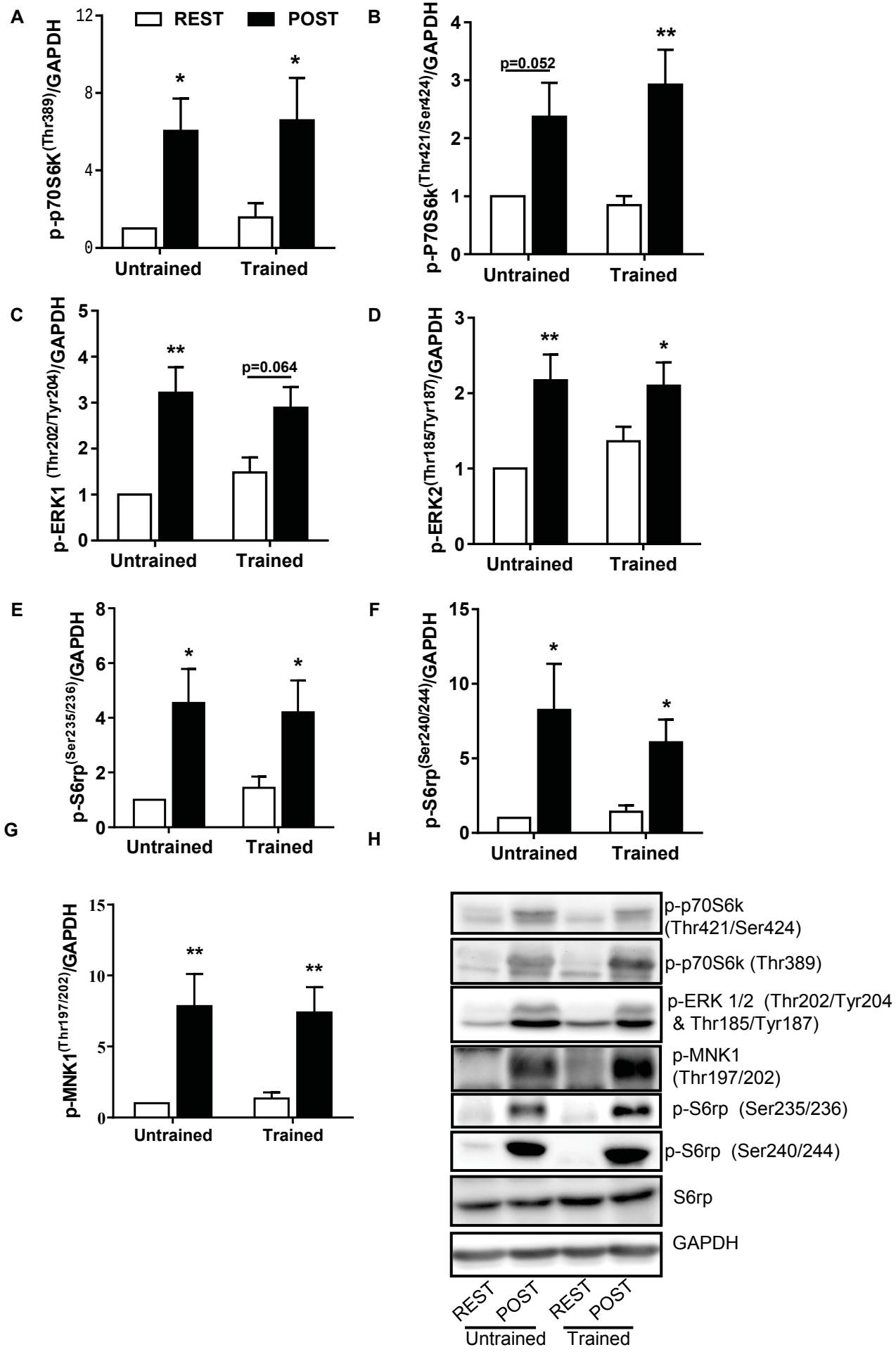

A
REST
POST
p-p70S6K(Thr389)/GAPDH
12
8
4
0
*
*
Untrained
Trained
B
p-P70S6k(Thr421/Ser424)/GAPDH
4
3
2
1
0
p=0.052
**
Untrained
Trained
C
p-ERK1 (Thr202/Tyr204)/GAPDH
5
4
3
2
1
0
**
p=0.064
Untrained
Trained
D
p-ERK2(Thr185/Tyr187)/GAPDH
3
2
1
0
**
*
Untrained
Trained
E
p-S6rp(Ser235/236)/GAPDH
8
6
4
2
0
*
*
Untrained
Trained
F
p-S6rp(Ser240/244)/GAPDH
15
10
5
0
*
*
Untrained
Trained
G
p-MNK1(Thr197/202)/GAPDH
15
10
5
0
**
**
Untrained
Trained
H
p-p70S6k (Thr421/Ser424)
p-p70S6k (Thr389)
p-ERK 1/2 (Thr202/Tyr204 & Thr185/Tyr187)
p-MNK1 (Thr197/202)
p-S6rp (Ser235/236)
p-S6rp (Ser240/244)
S6rp
GAPDH
REST
POST
REST
POST
Untrained
Trained

Figure 5.

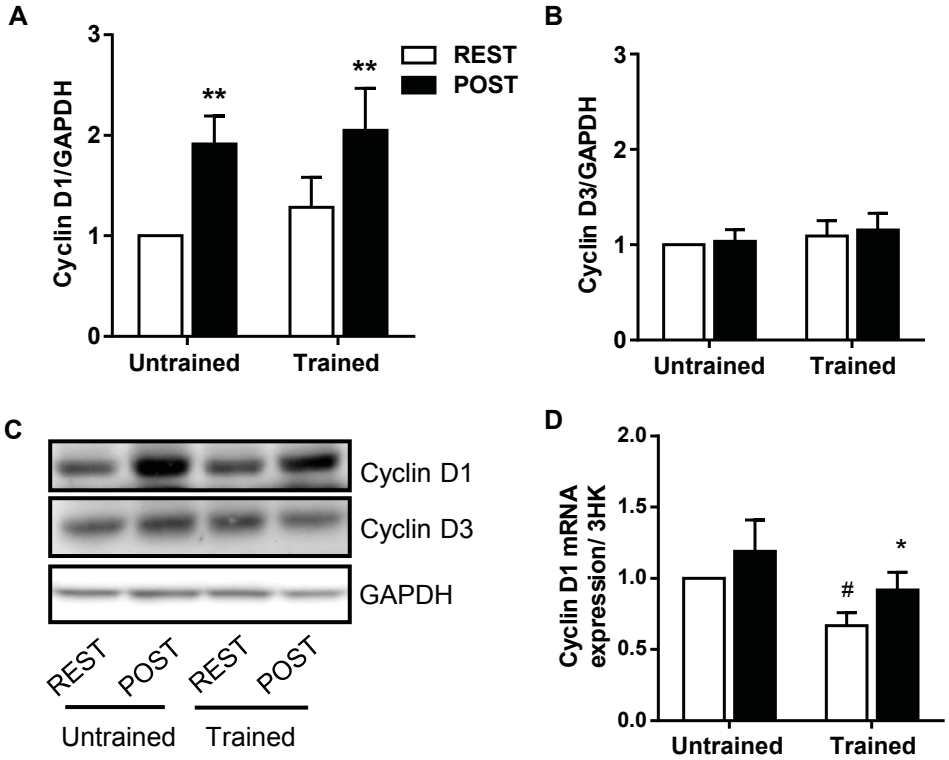

A
Cyclin D1/GAPDH
REST
POST
**
**
Untrained
Trained
B
Cyclin D3/GAPDH
Untrained
Trained
C
Cyclin D1
Cyclin D3
GAPDH
REST
POST
REST
POST
Untrained
Trained
D
Cyclin D1 mRNA expression/ 3HK
#
*
Untrained
Trained

Figure 6.

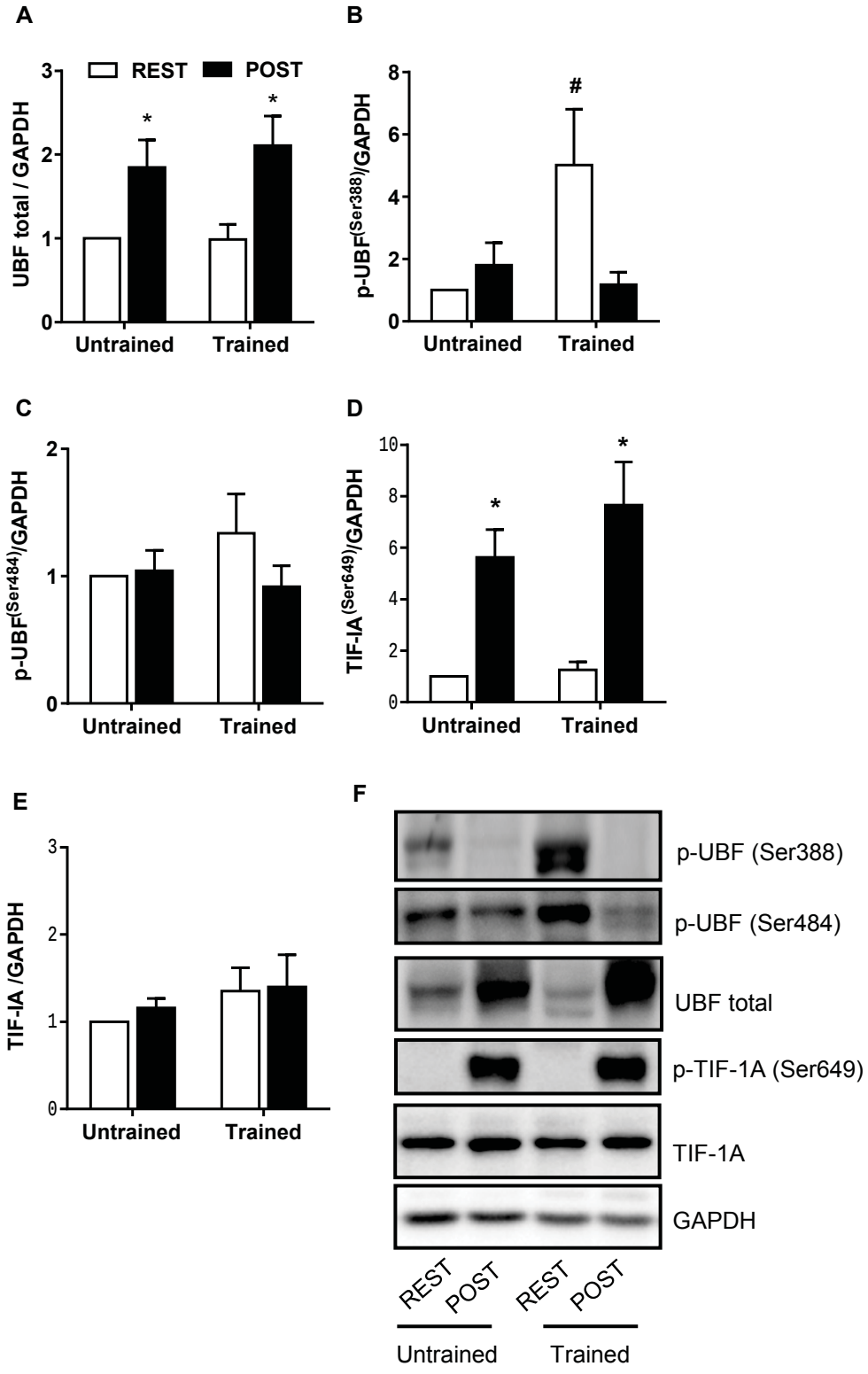

A
REST
POST
UBF total / GAPDH
Untrained
Trained
B
p-UBF(Ser388)/GAPDH
Untrained
Trained
C
p-UBF(Ser484)/GAPDH
Untrained
Trained
D
TIF-IA (Ser649)/GAPDH
Untrained
Trained
E
TIF-IA /GAPDH
Untrained
Trained
F
p-UBF (Ser388)
p-UBF (Ser484)
UBF total
p-TIF-1A (Ser649)
TIF-1A
GAPDH
REST
POST
REST
POST
Untrained
Trained

Figure 7.

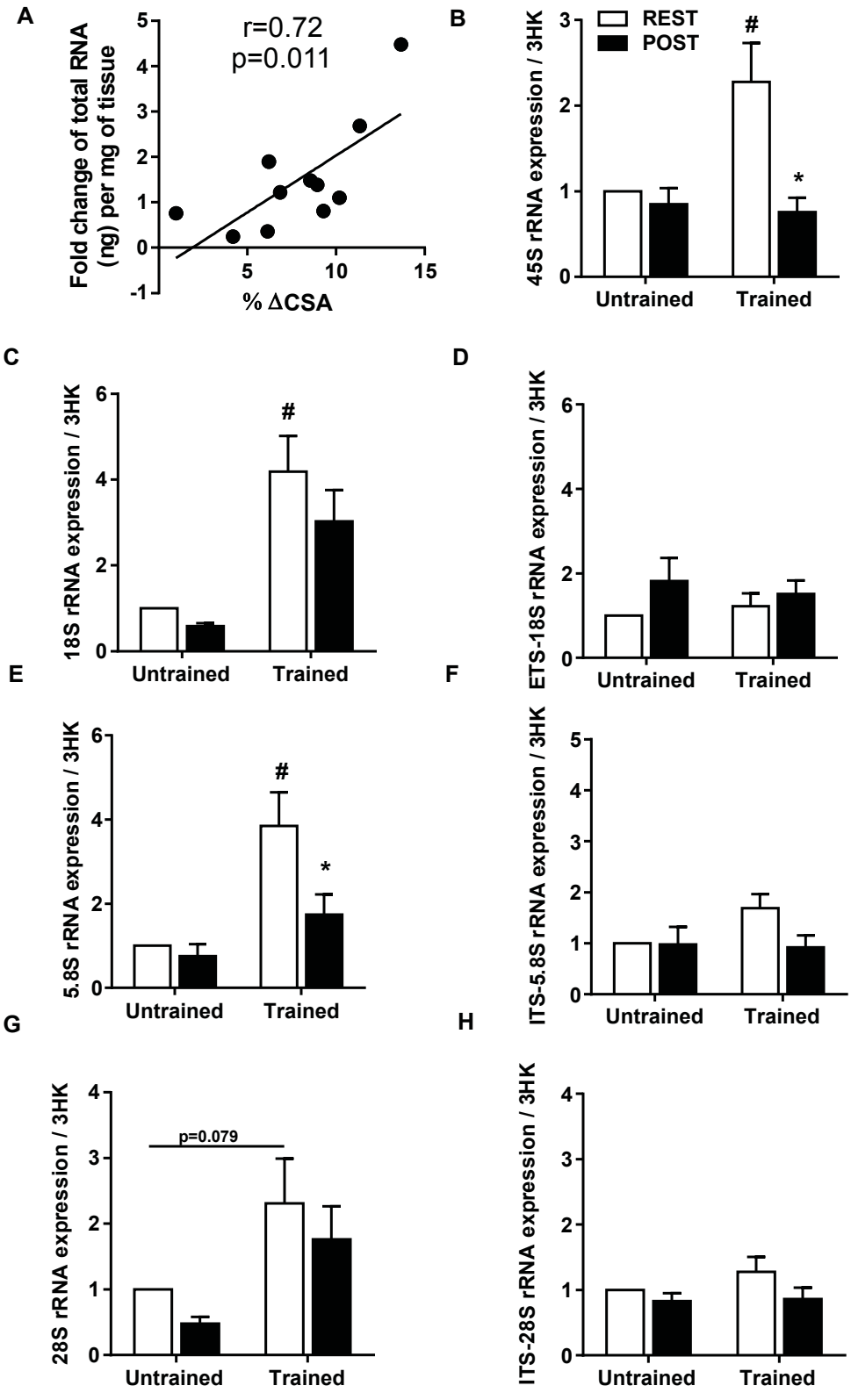

A
r=0.72
p=0.011
Fold change of total RNA (ng) per mg of tissue
% ΔCSA
B
REST
POST
45S rRNA expression / 3HK
Untrained
Trained
C
18S rRNA expression / 3HK
D
ETS-18S rRNA expression / 3HK
E
5.8S rRNA expression / 3HK
F
ITS-5.8S rRNA expression / 3HK
G
28S rRNA expression / 3HK
p=0.079
H
ITS-28S rRNA expression / 3HK

Figure 8.

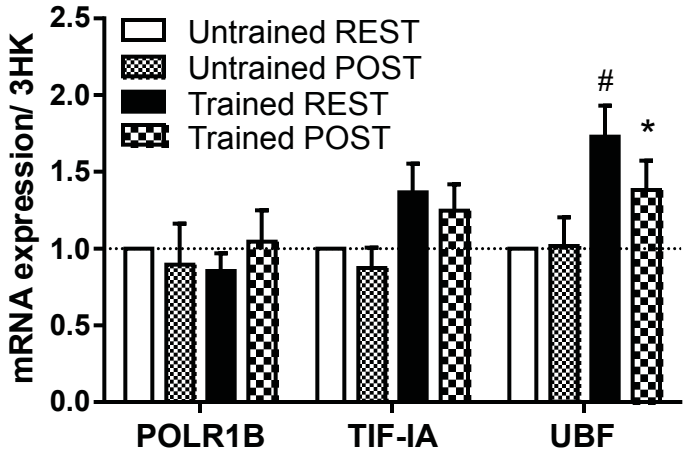
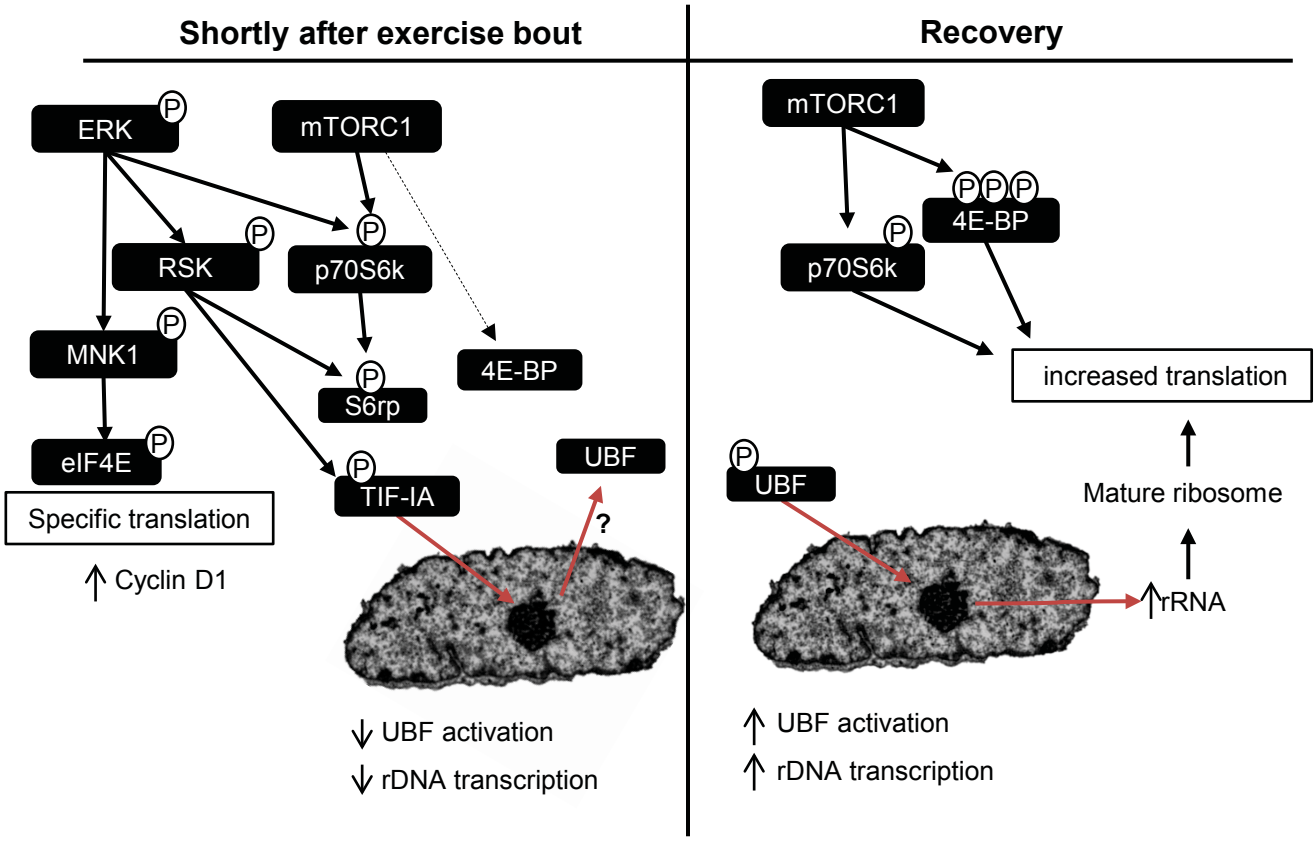

2.5
2.0
1.5
1.0
0.5
0.0
mRNA expression/ 3HK
Untrained REST
Untrained POST
Trained REST
Trained POST
#
*
POLR1B
TIF-IA
UBF

Figure 9.


Shortly after exercise bout
Recovery
ERK
mTORC1
RSK
p70S6k
MNK1
4E-BP
S6rp
eIF4E
TIF-IA
UBF
Specific translation
Cyclin D1
?
UBF activation
rDNA transcription
mTORC1
4E-BP
p70S6k
increased translation
Mature ribosome
UBF
rRNA
UBF activation
rDNA transcription